Exercise Sheet 17

Discrete Mathematics by Hongfei Fu, 2023.12.6

Name:______________ Student ID:______________ Class: ___________________

1. Show that a simple graph is a tree if and only if it is connected but the deletion of any of its edges produces a graph that is not connected.

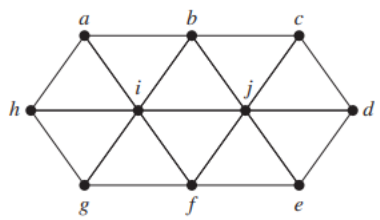
   Answer Area:

2. (**2pt**) Let G be a simple graph with n vertices. Show that G is a tree if and only if G has no simple circuits and has $n-1$ edges. [Hint: To show that G is connected if it has no simple circuits and $n-1$ edges, show that G cannot have more than one connected component.]

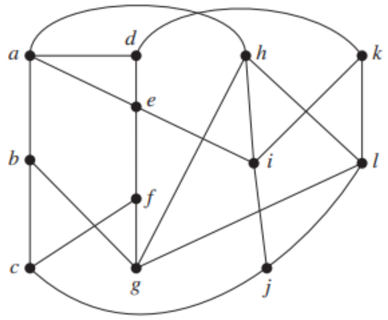
   Answer Area:

3. Find a spanning tree of the following graphs.

   a)



   b)



Answer Area: